

CSCI212 Lab — Quiz #4

Full Name:

EMPLID:

Date:

Score:

Below is **Java Swing** code written to execute a **GUI** application.

```
1 import javax.swing.*;
2 import java.awt.*;
3
4 public class Quiz4GUI {
5
6     private JPanel buttonPanel_1;
7     private JPanel buttonPanel_2;
8     private JLabel label_1;
9     private JLabel label_2;
10    private JTextArea textArea;
11
12    public Quiz4GUI() {
13
14        JFrame frame = new JFrame();
15        frame.setLayout(new BorderLayout());
16
17        buttonPanel_1 = new JPanel();
18        buttonPanel_2 = new JPanel();
19
20        textArea = new JTextArea(10, 30);
21
22        JButton B1 = new JButton("B1");
23        JButton B2 = new JButton("B2");
24        JButton B3 = new JButton("B3");
25        JButton A1 = new JButton("A1");
26        JButton A2 = new JButton("A2");
27        JButton A3 = new JButton("A3");
28
29        label_1 = new JLabel("L#1");
30        label_2 = new JLabel("L#2");
31
32        buttonPanel_1.add(B1);
33        buttonPanel_1.add(B2);
34        buttonPanel_1.add(B3);
35        buttonPanel_2.add(A1);
36        buttonPanel_2.add(A2);
37        buttonPanel_2.add(A3);
38
39        frame.add(buttonPanel_2, BorderLayout.NORTH);
40        frame.add(buttonPanel_1, BorderLayout.SOUTH);
41        frame.add(textArea, BorderLayout.CENTER);
42        frame.add(label_1, BorderLayout.WEST);
43        frame.add(label_2, BorderLayout.EAST);
44
45        frame.setSize(500, 500);
46        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
47        frame.setVisible(true);
48    }
49
50    public static void main(String[] args) {
51        new Quiz4GUI();
52    }
53 }
```

1. In the space below, sketch a schematic of the GUI defined in *Quiz4GUI* that properly positions the location of components defined in the constructor. Make sure to include the text titles corresponding to each component e.g. "B1" or "L#1", and use squares/rectangles to represent each component on the GUI. You do not need to label the position/location of the component relative to the frame, e.g. "NORTH", "EAST", ect. **The frame does not need to be sketched to scale, but is required to be legible enough to grade!**

2. **Subjective question (no wrong answers!)** Briefly discuss 1 or 2 GUI applications/projects you would like to see/work on in lab using **Java Swing**. Think about the types of user interfaces (UIs) or graphical menus you have seen in the past, and which ones you would like to recreate or model using swing. Get creative!